

Course Code: CS-3001	Course Name: Computer Networks
Instructor Names: Dr. Sufian Hameed, Dr. Farrukh Salim, Mr. Shoaib Raza, Ms. Yusra Kaleem	
Student Roll No:	Section No:

Instructions:

- Return the question paper.
- **All questions must be answered in answer script and according to the sequence given in the question paper.**
- In case of any ambiguity, you may make an assumption. But your assumption should not contradict any statement in the question paper.

Time: 60 minutes.

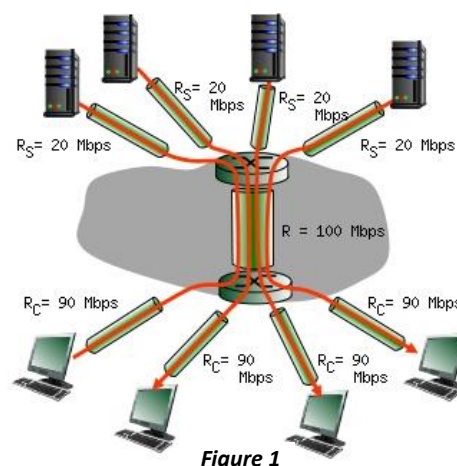
Max Marks: 50 Points

Question 1: Briefly discuss the following:

- What are the four different sources of packet delays? How does packet loss occur? **CLO#: 01 [05 Points]**
- List five layers, from top to bottom, in the Internet protocol stack? How many layers are implemented in a switch? **CLO#: 01 [05 Points]**

Question 2: Consider the scenario shown in figure 1, with four different servers connected to four different clients over four three-hop paths. The four pairs share a common middle hop with a transmission capacity of $R = 100$ Mbps. The four links from the servers to the shared link have a transmission capacity of $R_s = 20$ Mbps. Each of the four links from the shared middle link to a client has a transmission capacity of $R_c = 90$ Mbps per second.

CLO#: 01 [10 Points]



- What is the maximum achievable end-end throughput (in Mbps) for each of four client-to-server pairs, assuming that the middle link is fair-shared (i.e., divides its transmission rate equally among the four pairs)?
- Which link is the bottleneck link for each session?
- Assuming that the senders are sending at the maximum rate possible, what are the link utilizations for the sender links (R_s), client links (R_c), and the middle link (R)?

Question 3: The text below shows the reply sent from the server in response to the HTTP GET message. Answer the following questions, indicating where in the message you find the answer. **CLO#: 02 [10 Points]**

```

HTTP/1.1 200 OK<cr><lf>Date: Tue, 07 Mar 2008 12:39:45GMT<cr><lf>Server: Apache/2.0.52 (Fedora)
<cr><lf>Last-Modified: Sat, 10 Dec2005 18:27:46 GMT<cr><lf>ETag: "526c3-f228a4c80"<cr><lf>Accept-
Ranges: bytes<cr><lf>Content-Length: 3874<cr><lf>Keep-Alive: timeout=max=100<cr><lf>Connection: Keep-
Alive<cr><lf>Content-Type: text/html; charset=ISO-8859-1<cr><lf><cr><lf><!doctype html public "-//w3c//dtd html 4.0
transitional//en"><lf><html><lf><head><lf> <meta http-equiv="Content-Type" content="text/html; charset=iso-8859-
1"><lf> <metaname="GENERATOR" content="Mozilla/4.79 [en] (Windows NT5.0; U) Netscape]"><lf> <title>CMPSCI
453 / 591 / NTU-ST550A Spring 2005 homepage</title><lf></head><lf><much more document text following here (not
shown)>

```

- Was the server able to successfully find the document or not? What time was the document reply provided?
- When the document was last modified?
- How many bytes are there in the document being returned?
- What are the first 5 bytes of the document being returned? Did the server agree to a persistent connection?

Question 4: **nu.edu.pk** is registered and hosted with Dreamhost. Both the webserver and mailserver of **nu.edu.pk** are associated with 111.112.113.114 and 111.112.113.115 respectively. The primary authoritative name serve of Dreamhost is dns1.dreamhost.com which is mapped to 192.1.1.4. List down all the resource records (RRs) that will be inserted into the authoritative name server and .com's TLD (top level domain) server. **CLO#: 02 [10 Points]**

Question 5: For the result of the traceroute command shown below in Figure 2, give answers for below questions:

CLO#: 01 [10 Points]

- If traceroute command runs again, will the IP address shown by point A will be always this same for google.com? If no, and any other IP address is also possible, then why?
- What this '30 hops' represents?
- Which device is represented by the IP address of Point C in the network?
- What * represents, as shown by point E?
- On running traceroute command, the result for this line shown by point E is always 'Request timed out'. What could be the possible reason behind this?
- What these three different values represents, as shown by point F or G, and why these three values are mostly not same? And why not only one value suffices?
- Why the values of F (hop 7) are mostly smaller than values of G (hop 8)? Give reason.
- What is the importance of this number 13 shown by point H?
- What this URL is, as shown by point I?

```
C:\Users\Public>tracert google.com

Tracing route to google.com [172.217.19.174] A
over a maximum of 30 hops: B
  0  1 ms    1 ms    1 ms    192.168.43.1 C
  1  D *      *      *      Request timed out. E
  2  35 ms   49 ms   44 ms   10.81.77.165
  3  51 ms   36 ms   49 ms   10.81.73.137
  4  179 ms  50 ms   47 ms   10.180.76.1
  5  40 ms   75 ms   70 ms   10.81.171.21
  6  522 ms  67 ms   138 ms  116.71.129.225
  7  F 60 ms   36 ms   40 ms   10.253.4.44
  8  G 58 ms   78 ms   65 ms   10.253.4.8
  9  89 ms   86 ms   90 ms   74.125.118.170
 10  79 ms   55 ms   58 ms   172.253.51.205
 11  57 ms   56 ms   61 ms   209.85.250.17
 12  H 56 ms   57 ms   55 ms   zrh04s07-in-f174.1e100.net [172.217.19.174] I
Trace complete.
```

Figure 2

Best of Luck!!!